Industrial Visual Anomaly Detection: A Technical Case Study

Extended brief (two pages) · independent project on public benchmark images (VisA), not factory data

Programmatic metric verification completed: a separate implementation recomputed the reported results from the recorded per-image scores. AI-assisted technical review performed; human owner review and acceptance pending. The one-page summary is the default executive version.

The problem

Can a model fitted on good boards flag defects without excessive false alarms? For the normal-only pipeline, VisA defect labels and masks are reserved for evaluation. The separate PCB4 supervised study uses defect annotations for YOLO training.

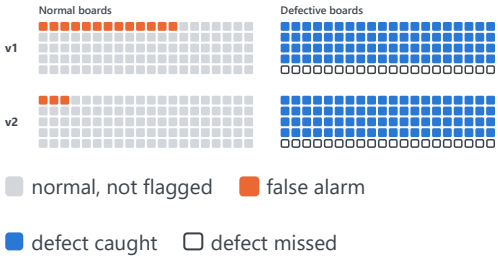
What was built

A working prototype: frozen pretrained CNN features, a per-position Gaussian (Mahalanobis) model of normal appearance, board registration and — in v2 — a score taken over the board region only. Thresholds are frozen on separate normal images before testing; each controlled change was pre-registered and evaluated once on a board type not used in its development, with deviations disclosed.

Measured result — held-back board type PCB2

100 defective + 100 normal test images; 95% confidence intervals

	v1	v2	Change
Defect recall	80%	**80%**	unchanged
False alarms, test normals	13%	**3%**	−10 pts (CI −17 to −3)
False alarms, holdout	5.3%	**4.0%**	−1.3 pts (CI −4.6 to +2.0)
AUROC	0.927	**0.979**	+0.052 (CI +0.028 to +0.082)



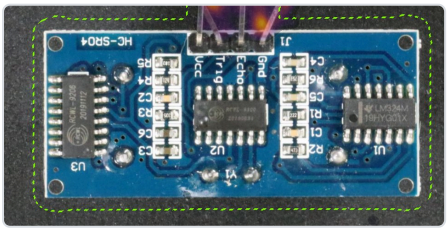
Median end-to-end latency per image (GPU, one image at a time): v1 194 ms, v2 224 ms. Although v2 improved false-positive performance, it did not meet the proposed detection-sensitivity and robustness requirements. It is not ready for production-affecting deployment. Recall stays below the proposed 90% target; rule-based verdict against the unchanged criteria: INCONCLUSIVE.

Remaining technical risks

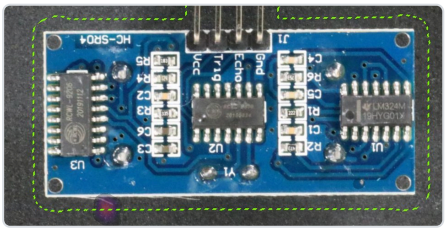
- **Detection.** Recall 80% against a proposed 90%; the smallest third of defects only 56%.
- **Lighting.** A 10% darker image raised PCB2 false alarms from 3.6% to 28.7%.
- **Portability.** On PCB3's green background the original board-finding rule marked 96% of the frame as board, silently.
- **Board region.** About 31% of the PCB3 scoring region is background cloth; lint near the board can still raise a false alarm.

RECOMMENDATION

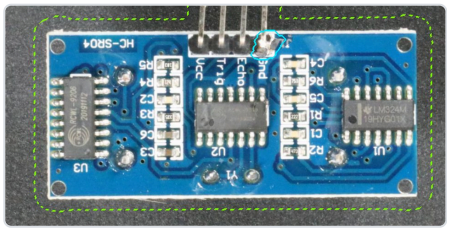
Not ready for production-affecting deployment. The proposed next step is a bounded, non-disruptive shadow-mode validation, subject to site requirements, authorization and appropriate entry checks. Factory access has not been obtained.



Defect caught: score 88.6 vs threshold 51.2



False alarm (peak on cloth near board): score 65.1 vs threshold 51.2



Small defect missed (cyan = ground truth): score 31.4 vs threshold 51.2

Follow-up experiment: lighting correction on a new board type (PCB3)

One pre-registered change — standardise each image's global brightness and contrast before registration — compared with the unchanged pipeline on 256 unseen normal and 100 defective images, evaluated once.

Condition	False alarms, baseline	False alarms, corrected	Role
Clean	4.3%	**3.5%**	reference; recall baseline 90/100, corrected 91/100
Global darkening (G1)	8.6%	**3.1%**	primary endpoint
Brightening (B1)	19.5%	**6.2%**	secondary
Non-uniform lighting (N1)	6.2%	**5.9%**	secondary, out of class
Mild blur	4.7%	**18.0%**	acceptance check only

- **A. Pre-registered outcome:** "ROBUSTNESS IMPROVEMENT SUPPORTED" (darkening difference −4.7 pts, CI −8.6 to −0.8) — **weakly informative**: the baseline showed only a small false-alarm increase under darkening, below the protocol's informativeness threshold.
- **B. Engineering decision: do not adopt the correction.** It made a mild blur raise false alarms by +14.5 pts (uncorrected +0.4), beyond the original 10-pt limit. Non-uniform lighting was not improved; part of the effect is likely more stable registration.
- **The baseline is not production-ready either:** brightening raised its false alarms by +15.2 pts; both variants are INCONCLUSIVE against the unchanged criteria.
- **Not a progression:** PCB1, PCB2 and PCB3 are separate board types with their own models and thresholds. The PCB3 baseline's 90% recall is not comparable with PCB2's 80% and does not predict recall on a real line.
- **Disclosure:** after PCB3 test scoring, one input check in the analysis program was repaired (7dbd302); no score, threshold or decision changed, and a separate verification script agrees on 90/90 checks (repository: docs/pcb3_post_evaluation_repair.md).

Separate exploratory study: PCB4

YOLOv8s and YOLO11s were actually trained on a separate PCB4 dataset/protocol. With different supervision and annotation budgets, they achieved higher observed image-level recall in the tested supervised setting. Brightness-related acquisition confounding and weakness on unseen defect types limit interpretation; image-level recall does not establish correct localization. Findings in the case study; detailed records remain in the private development repository.

Entry checks for proposed shadow validation

- Board segmentation and registration on site normal images.
- Lighting checks and a perturbation test at the site's frozen threshold.
- Site defect samples; recall reported by defect size.
- Review capacity, miss/review costs, and versioned model + threshold releases with rollback.

Engineering discipline — practices by status

Frozen thresholds, recorded scores, pre-registered changes and disclosed deviations support traceability. Team practices remain proposals; no human peer review, continuous integration or line operation. The website records each practice by status.

Not yet validated: Any factory image, camera, lamp, line speed or real defect mix. · Usability of the review interface; reviewer agreement. · Labour saving or financial return. · Robustness to real (not synthetic) lighting changes.

"ROI" here means the image region of interest. Production deployment needs real operational constraints and representative factory data before technical acceptance criteria can be validated. No financial return on investment or labour saving is claimed.

Data: VisA dataset, Zou et al., ECCV 2022, CC BY 4.0. Figures resized, registered, cropped, with overlays. PCB1–PCB4 are benchmark board types, not manufacturing sites. Not affiliated with or endorsed by the dataset authors or any company.

Built with AI-assisted development; project-owner technical review pending. Project owner: Yu-Chen (Leon) Lai.

Website link pending first deployment.